



Tropentag, September 16 - 18, 2026, Göttingen

"Towards multi-functional agro-ecosystems
promoting climate-resilient futures"

Agronomic and economic performance of integrated soil fertility management across an organic matter continuum and climate variability

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Abstract

We develop a hybrid dynamic Bayesian Network model to capture the probabilistic relationships among organic matter, soil fertility trajectories, climate variability, crop variety-specific responses, and socio-economic feasibility within a unified framework. We demonstrate the utility of this model architecture by applying it to smallholder systems in the Guinea Savannah of northern Ghana, where Integrated Soil Fertility Management (ISFM) recommends up to 5 tonnes ha⁻¹ of organic matter to restore fertility and sustain productivity on degraded lands. However, farmers relying on crop residues as primary organic inputs rarely achieve this rate due to competing uses of crop residues. Our model assesses the agronomic and economic performance of ISFM across an organic matter input continuum of 1-5 tonnes ha⁻¹ for locally relevant varieties of maize and soybean. For each organic matter input level, we assess how soil organic matter stocks change under current and projected climates, how these dynamics translate into crop productivity for each crop variety, and whether resulting yields are agronomically sufficient and economically viable given the labour and management costs of crop residue integration. Our results reveal that while 5 tonnes ha⁻¹ of organic inputs maximises absolute soil organic matter recovery, it is not the universal economic optimum for resource-constrained smallholders. Instead, our model identifies a threshold of diminishing returns, where the joint probability of achieving soil fertility recovery, agronomic sufficiency, and economic viability is maximised under climate variability. The work can guide extension and policy on pragmatic ISFM implementation. The model provides a decision-support framework tailored to real-world constraints. By aligning extension efforts with the dual goals of soil fertility restoration and economic viability, this methodology offers a robust, scalable framework for designing site-specific, resilient agricultural intensification pathways in the Guinea Savannah and beyond.

Keywords: Bayesian network, climate change, crop suitability, Monte Carlo simulation, soil organic matter

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